

SQL Interview Questions By Level

This document contains the most important SQL interview questions collected and organized from the provided PDFs. Questions are separated into Junior, Mid-Level, and Senior-Level categories.

Junior SQL Interview Questions

#	Interview Question
1	What is the difference between DELETE, TRUNCATE, and DROP?
2	What are Primary Key and Foreign Key?
3	What is the difference between WHERE and HAVING?
4	What is GROUP BY in SQL?
5	What is the difference between GROUP BY and ORDER BY?
6	What are aggregate functions in SQL?
7	What is a JOIN? Explain INNER JOIN, LEFT JOIN, RIGHT JOIN.
8	What is a Subquery (Nested Query)?
9	How do you find duplicate records in a table?
10	How do you find the second highest salary?
11	What is a CASE statement?
12	What is COALESCE in SQL?
13	What is Indexing in SQL?
14	Difference between Clustered and Non-Clustered Index.
15	What is a Temporary Table?

Mid-Level SQL Interview Questions

#	Interview Question
1	What is a Correlated Subquery?
2	What is a CTE (Common Table Expression)?
3	What are Window Functions in SQL?
4	Difference between ROW_NUMBER(), RANK(), and DENSE_RANK().
5	How do you calculate running totals using window functions?
6	How do you use LAG() and LEAD()?
7	How do you calculate moving averages in SQL?
8	How do you retrieve the latest record per customer?
9	How do you detect gaps and islands in SQL?
10	How do you identify overlapping date ranges?

11	How do you pivot rows into columns?
12	How do you unpivot columns into rows?
13	How do you rank employees by department salary?
14	How do you find customers with no purchases?
15	How do you calculate monthly sales growth percentage?

Senior SQL Interview Questions

#	Interview Question
1	Explain recursive CTEs with hierarchy traversal examples.
2	How do you detect circular references in employee-manager hierarchy?
3	How do you optimize slow SQL queries?
4	Explain execution plans and query optimization.
5	How do indexes affect INSERT, UPDATE, and DELETE performance?
6	How do you calculate median salary using window functions?
7	How do you build scalable reporting queries?
8	How do you handle very large datasets efficiently?
9	How do you write recursive queries for organizational hierarchy?
10	How do you identify top N products per month?
11	How do you calculate cumulative distribution using CUME_DIST()?
12	How do you calculate percentiles using PERCENTILE_CONT()?
13	How do you detect missing invoice numbers?
14	How do you compare two tables and find differences in all columns?
15	How do you design SQL queries for analytics and trend analysis?

Tip: Focus first on understanding concepts, then practice writing queries manually. Most interviewers expect clear explanations in addition to correct SQL syntax.